

VLSI DSP 2026 Homework Assignment #3

Due date: 2026/5/12

Q1. For the DG shown in Figure 1,

(a) Which of the following sets of scheduling and projection are permissible?

- i. $\mathbf{s} = [1 \ 0]^T$, $\mathbf{d} = [1 \ 0]^T$
- ii. $\mathbf{s} = [1 \ 2]^T$, $\mathbf{d} = [2 \ -1]^T$
- iii. $\mathbf{s} = [1 \ 1]^T$, $\mathbf{d} = [1 \ 0]^T$
- iv. $\mathbf{s} = [1 \ -2]^T$, $\mathbf{d} = [1 \ 0]^T$
- v. $\mathbf{s} = [2 \ 1]^T$, $\mathbf{d} = [1 \ 0]^T$

(b) derive the projected systolic array for each permissible set

(c) extend the DG to a 5-tap linear phase filter and derive its mapping subject to the following two constraints 1) pipeline period = 1, 2) critical path delay is no greater than 1 multiplier plus two adders

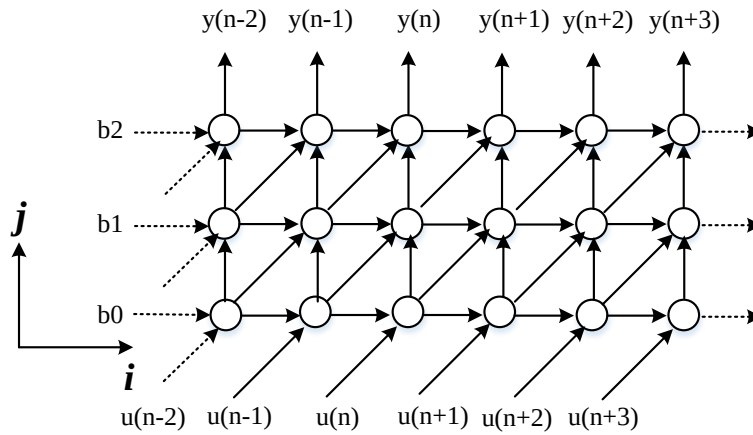


Figure 1

Q2. Vector quantization design

Given an input k -dimensional column vector $\mathbf{r}_{k \times 1}$, and a codebook $\mathcal{B} = \{\mathbf{b}_i \mid i = 1 \sim N\}$ consists of N k -dimensional column vectors, vector quantization (VQ) is to find a vector in that codebook that has the shortest Euclidean distance from the input vector $\mathbf{r}_{k \times 1}$. The Euclidean distance between two vectors (also called square error) is defined as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|^2 = \sum_{j=0}^{k-1} (x_j - y_j)^2 \quad (1)$$

Let $\mathbf{x} = \mathbf{r}$ and $\mathbf{y} = \mathbf{b}_i$, Eq(1) can be rewritten as $d(\mathbf{r}, \mathbf{b}_i) = \|\mathbf{r}\|^2 - 2\mathbf{r}^T \mathbf{b}_i + \|\mathbf{b}_i\|^2$. Since $\|\mathbf{r}\|^2$ is a constant term in all Euclidean distance calculation, and $\|\mathbf{b}_i\|^2$ can be precomputed, VQ calculation can be expressed as

$$\arg\{\min\{c_i - \mathbf{r}^T \mathbf{b}_i \mid i = 1, N\}\} \quad (2)$$

Where $c_i = \|\mathbf{b}_i\|^2 / 2$. In other words, VQ can be accomplished by searching the entire codebook and for each codebook vector $\mathbf{b}_i$, the value $c_i - \mathbf{r}^T \mathbf{b}_i$ is computed and the smallest one is recorded. Note that $\mathbf{r}^T \mathbf{b}_i$ is an inner product operation, and c_i can be input from a pre-computed table.

(a) Please draw the DG of the $y(i) = c_i - \mathbf{r}^T \mathbf{b}_i$ for $i = 1 \sim N$. For simplicity, assume the vector dimension k is 3. In each iteration, c_i and $\mathbf{b}_i$ are regarded as input and $y(i)$ is the output.

- (b) Select a scheduling and projection scheme to obtain its systolic array design (at least one delay element in every inter-processor data link).
- (c) Add a comparator module so that the vector index i corresponding to the minimum Euclidean distance can be obtained at the end of N iterations.